

CS 2XY3 - Problem Set 2: Memory Management

Due: Week 6. Submit via your LMS.

Question 1 (20 points)

Consider a logical address space of 256 pages with a 4-KB page size, mapped onto a physical memory of 128 frames.

- (a) How many bits are required for the logical address?
- (b) How many bits are required for the physical address?
- (c) What is the size of the page table if each entry contains the frame number plus 3 protection bits?

Question 2 (15 points)

Explain the difference between internal fragmentation and external fragmentation. Which type of fragmentation does paging eliminate? Which does segmentation suffer from?

Question 3 (15 points)

A system uses a two-level page table. The virtual address is 32 bits, page size is 4 KB, and each page table entry is 4 bytes. Calculate: (a) the number of entries in the inner page table, (b) the number of entries in the outer page table, (c) the total memory overhead if only 3 inner tables are actually used.